

the core shroud 22 will also tend to reduce core power by reducing the core flow and will not cause the power to increase by reducing the temperature of water entering the core from the downcomer. Core power can be regulated by regulating the fraction of feedwater flow that is directed inside the core shroud 22 via one or more alternative supply lines 52. This may occur with or without the operation of the isolation condenser system 30. The feedwater, like the isolation condenser system 30 condensate, may be distributed by sparger piping as is known in the art.

[0036] Simulation results are presented for a natural circulation boiling water reactor accident scenario of an anticipated transient without scram. The transient scenario begins with abrupt closure of the turbine stop valve at one second which would cause a scram signal. The mechanisms for hydraulic scram are assumed to fail so the only remaining way to scram is for the control rod drive motors to drive the control blades into the core. This scram is significantly slower than the hydraulic scram. The scram speed is constant with linear insertion of the control blades beginning at the turbine stop valve trip signal plus an assumed short electronic delay. The scram time is defined as the elapsed time for a controlled blade that is initially fully withdrawn from the core to become fully inserted. In the simulations, the condensate return valves 54 for the specified number of isolation condensers 30 also began to open at the turbine stop value trip signal and complete their stroke linearly to full open in 10 seconds.

[0037] The simulation results are divided into 3 sets. The first set of results are depicted in Figure 2 through Figure 5. These results all have a 30 second scram time so that the same basis can be used for comparing